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0.1 Cleaning Initial Dataset

- Initial Dataset from US Department of Transportation titled “*Monthly Modal Time Series*”
- Filtered to California Specific Data for higher relevance. Filter Specifically to your state or area for a dataset that’s easier for students to explore and model.

```
library(readr)
library(tidyverse)
library(stringr)
pt_data <- read_csv("NTD_TimeSeries.csv")

pt_data1 <- pt_data|>
  filter(str_detect(Primary.UZA.Name, "CA$"))

write_csv(pt_data1, "ca_transit.csv")
```

0.2 Dataset: National Transit Database (NTD)

Discuss: The goal here is to get students to understand that the NTD dataset itself does not contain enough information for the insightful analysis we are looking for.

0.3 Cleaning Distribution Materials

```
library(sf)
library(tigris)
library(tidyverse)

# Original sf at California Open Data Portal titled "California Transit Stops"
cts_sf <- st_read("cts_sf")
```

```

# pulling geography from Tigris and transforming to uniform CRS (NAD83)
ca_tracts <- tracts(state = "CA", year = 2024, class = "sf")
stops_sf <- st_transform(cts_sf, crs = st_crs(ca_tracts))

# Joining where stop is within tract
stops_with_tract <- st_join(stops_sf, ca_tracts, join = st_within)

# Separate matched and unmatched stops
matched_stops <- stops_with_tract |> filter(!is.na(GEOID))
unmatched_stops <- stops_with_tract |> filter(is.na(GEOID))

# match unmatched stops
nearest_idx <- st_nearest_feature(unmatched_stops, ca_tracts)

# Pull the nearest GEOIDs from the tracts
unmatched_stops_fixed <- unmatched_stops |>
  mutate(
    GEOID = ca_tracts$GEOID[nearest_idx],
  )

clean_sf <- bind_rows(matched_stops, unmatched_stops_fixed) |>
  select(agency, stop_id, stop_name, routetypes, n_arrivals, n_hours_in,
    district_n, GEOID, geometry)

# Creates sf folder in working directory
dir.create("output_shapefile")
st_write(clean_sf, "output_shapefile/CA_Transit_Stops.shp", delete_layer = TRUE)

```

Retrieving census data from tidycensus package

```

# Pulling relevant variables
vars <- c(median_income = "B19013_001",
  num_pov = "B17001_002",
  avg_commute = "DP03_0025",
  no_vehicle = "B08201_002",
  white = "B02001_002",
  black = "B02001_003",
  asian = "B02001_005",
  hispanic_latino = "B03003_003",
  transit_users = "B08301_010",
  tot_workers = "B08301_001",
  total_pop = "B02001_001"
)

# Retrieve unique API here "https://api.census.gov/data/key_signup.html"
library(tidycensus)

```

```

census_data <- get_acs(
  geography = "tract",
  variables = vars,
  state = "CA",
  year = 2023, # Most recent acs5 survey
  survey = "acs5",
  geometry = FALSE)

# Cleaning and Transforming Data
census_data1 <- census_data |>
  select(GEOID, variable, estimate)|>
  pivot_wider(names_from = variable, values_from = estimate) |>
  mutate(pctpoc = ((black + hispanic_latino + asian ) / total_pop) * 100)|>
  mutate(pctpov = (num_pov / total_pop) * 100)|>
  mutate(pct_wo_car = (no_vehicle / total_pop) * 100)|>
  mutate(pct_transit_users = 100 * (transit_users / total_pop))|>
  mutate(GEOID = as.character(GEOID))|>
  select(-black, -asian, -white, -hispanic_latino, -no_vehicle, -num_pov, -transit_users)

# Join with Census Tracts Geomotry

census_data2 <- census_data1|>
  right_join(ca_tracts, by = "GEOID")

dir.create("output_shapefile2")
st_write(census_data2, "output_shapefile2/CA_Census_Data.shp", delete_layer = TRUE)

```

0.4 Lesson Materials

The worksheet below is intended to follow along with the “**Public Transit Lesson**” lecture slides. With this worksheet specifically, students should also have a computer, or be in groups with a computer, so they can practice some data exploration and analysis in real time.

In-Class Data Wrangling Practice

0.5 Interactive ggplot

In the section of the lesson “Visualizing with ggplot2”, students are prompted to work in groups or pairs to come up with the code to create a specific line plot. In this section, it may be beneficial to have rstudio or your R coding interface open and projected so students can follow along in the coding process. **For example**, the first question in the worksheet asks students to tweak a previous wrangling example and then plot their findings. Project the code below or write it on the board, and once students have finished discussing in groups, have them

help to fill in the blanks and complete the wrangling task together.

```
summary |>
  filter(Agency == "____") |>
  ggplot(aes(x = ___, y = ___)) +
  ___(linewidth = 0.6) +
  labs(
    title = "____",
    y = "____",
    x = "____"
  )
```

0.6 Mathlinks Lesson on Joining

In the previous lesson(Tree Equity), students used Mathlink Cubes to learn basic data wrangling principles. The same activity can be applied to learn the concept of joining datasets. Below are a few images that visualize the process of different join functions with the mathlink cubes. Showing these images as well as hands on interaction with the cubes are great ways to help cement the new data wrangling concept. Refer back to the “Tree Equity Lesson” for more information about how to best set this activity up in the classroom.

right_join

left_join

full_join

0.7 Joined Data

Introduce students to the new dataset. `census_data` has variables such as `pct_pov` or `pct_wo_car`. These variable can help us commit to a better understanding of how the transit stop data is affected by/affects society.

When discussing **`right_join()`** vs. **`left_join()`**, first have students make observations about the resulting datasets when using the functions on `stop_data`.

- Point out the Variables in each resulting dataset
- Point out the number of observations in each dataset
- notice how using right join leaves us with empty geometries/tracts (is this something we want to visualize?)

0.8 Exploring Relationships

In this section, guide students through wrangling and plotting exercises to look at the relationships between the transit data in an area and the socioeconomic markers in that area.

Goal: Allow students to come up with their own path to answer each question.

- **Provide** time for students to work on each problem in pairs/groups
- **Use** skeleton/scaffolded R-code to ease the transition from looking at code to writing it alone
- **Guide** students back to the dataset if their are unsure of variable names and values
- **Allow** different students to answer questions and explain their problem solving techniques to the class

0.8.1 Exploration Question Scaffolds

Q: How does pctpov (percent in poverty) relate to the number of transit stops in a neighborhood?

```
# Wrangling to Count Stops per Tract

full_data <- full_data |>
  group_by(____)|>
  ____ (stop_count = n())

full_data|>
  ggplot(aes(x = pctpov, y = ____)) +
  ____ (color = "lightgreen") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "____",
        x = "____",
        y = "____") +
  theme_grey()
```

Q: Do higher-income areas have more or fewer transit stops?

```
full_data|>
  ggplot(aes(x = ____, y = ____)) +
  ____ (color = "lightblue") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "____",
        x = "____",
        y = "____") +
  theme_grey()
```

Q: Do areas where more people don't own cars have better transit coverage?

```
plot_data|>
  ggplot(____(x = ____, y = ____)) +
  ____ (color = "yellow") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "____",
```

```

    x = "___",
    y = "___") +
  theme_grey()

```

0.9 Culminating Ideas

Throughout this entirety of this lesson, the goal has been to show students how to convey the most insightful findings based on their data. Through scatter plots, students can see a correlative relationship between 2 variables, but the spatial heat map allows us to visualize an extra layer of the puzzle: location. **Explain** to students how location is another variable that can make our conclusions about data more insightful and lead us to more educated action.

```

# Make adjustments to heat map here

facet_data <- full_data|>
  pivot_longer(cols = c(pctpov, pctpoc, median_income, pct_wo_car),
               names_to = "census_var", values_to = "value")

facet_data|>
  ggplot() +
  geom_sf(aes(fill = value), color = NA) +
  scale_fill_viridis_c() +
  facet_wrap(~ census_var, ncol = 1) +
  theme_minimal()

```